

Xiang (Shawn) Zhang

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<https://shawnzhg.github.io/>

Address: Ann Arbor, MI, USA

EDUCATION

University of Michigan, Ann Arbor

United States

Major: M. S. in Data Science

08/2025-06/2027

Average GPA: **3.9**

Core Courses: Machine learning (A), Web System.

Capital University of Economics and Business

Beijing, China

Major: B. S. in Data Science

09/2021-07/2025

Average GPA: **4.0**

Core Courses: Natural language processing, Operational research, Advanced Algebra, Artificial Intelligence, Data Mining.

RESEARCH INTERESTS

Where a language model's task knowledge and computation should live — outside the prompt, outside the weights — and how RL credit for it is assigned. External loadable capabilities, credit assignment in RLVR, latent-space reasoning.

RESEARCH EXPERIENCE

LLM Post training and Latent Reasoning

Ann Arbor, USA

Research Assistant at University of Michigan Liu lab (Jie Liu)

01/2026-05/2026

- **Literature-writing agent** (multi-turn reward design for **RL post training**): The writing task has no binary reward (unlike math/coding) and GRPO's single terminal reward barely propagates across a long generation, designed rubric-grounded, multi-turn, multi-dimensional rewards across the full trajectory.
- **Biomedical deep-research agent**: To curb hallucination and citation fabrication in literature review (where reliability is non-negotiable), co-designed a training-free, **multi-module agent** (compress-before-reason and verbatim-grounded full-text hybrid answers enforced by a mechanical faithfulness gate), raising fact-level completeness 0.442→0.645 and entity coverage 0.676→0.823 at 55% lower cost with zero fabricated citations.
(<https://github.com/shawnzhg/pubmed-deep-research-agent>)
- **KV-cache compression and reuse** (latent-space reasoning): Since natural-language reasoning chains scale token/context cost with length, compressed reusable "Skills" into a small set of KV-cache states and trained them as RL-optimizable parameters, which improve the accuracy of SearchQA from 77.7→88.3. Now the KV-Skill preprint (see below).
(These functions and agents will go live at <https://glkb.org> soon.)
- **Latent-space reasoning**: Designed MCIG, a training-free per-token redundancy signal, and a latent-feedback adapter (curriculum SFT + MCIG-guided RL, Qwen2.5-1.5B) to fold redundant reasoning into latent space; ablations isolated four mechanistic roadblocks (offline-discriminative-but-RL-useless signal, error accumulation, KV-cache poisoning/anchor collapse, RoPE-manifold mismatch) showing why pure latent skipping collapses at sub-2B scale (64.5%→55% at one latent step) and motivating selective latent↔explicit switching.
(<https://github.com/shawnzhg/adaptive-latent-reasoning/blob/main/REPORT.md>)

Reasoning Language Models underthinking (Post training of LLM)

Beijing, China / Remote

Research Assistant at Institute of Automation, Chinese Academy of Sciences(CASIA)

06/2025- 04/2026

- **Depth-Quality Erosion (RL training diagnosis)**: Suspecting GRPO/RLVR's rising aggregate Pass@1 masked real regressions, analyzed post-training at per-reasoning-depth granularity and derived a closed-form variance-update identity — uncovering a Simpson's-paradox mode-seeking collapse where pass rate drops in every depth bin (up to -26%) even as Pass@1 rises (+2.0 pp).
- **STAGE-GRPO**: To enrich the sparse group signal left by GRPO's terminal-only reward — the root of the erosion above — proposed a stratified-advantage variant with online per-prompt difficulty estimation that provably decouples variance dynamics from geometric alignment, eliminating the per-bin degradation and lifting MATH-500 Pass@1 to 0.868 (+4.2

pp vs. base, +2.2 pp vs. GRPO).

- **Prefix-contrastive credit assignment:** Observed GRPO spreads one terminal advantage across all tokens though only a few decide correctness (so the policy games length); designed a GRPO-identity-preserving per-token estimator that concentrates gradient on decision-critical tokens by contrasting, within a group, the branch positions where same-prefix correct/incorrect rollouts diverge; end-to-end ablations in progress.

(<https://github.com/shawnzhg/grpo-underthinking/blob/main/REPORT.md>)

Spatiotemporal Drought Forecasting via Transformer & GraphRAG

Beijing, China

Research Assistant, Chinese Academy of Agricultural Sciences(CAAS)

02/2025- 06/2025

- Proposed **Cross-Spatiotemporal-Former (CSTF)**: Designed a novel Transformer-based architecture with **Cross-Attention** by crossing the query of attention to fuse spatial (grid) and temporal (time-series) features, achieving 98.6% accuracy in drought classification.
- Outperformed Baselines: Validated on 12 years of ERA5-Land data (0.1 resolution); the model outperforming LSTM etc. baselines by capturing long-range dependencies in SPI/SPEI indices.
- Developed **GraphRAG System**: Constructed a domain-specific Knowledge Graph (Drought-Water-Wheat) using DistilBERT and FAISS; integrated with Baichuan2-13B to reduce LLM hallucinations and provide evidence-based agricultural solutions.

PUBLICATIONS & PREPRINTS

- Han, Z., **Zhang, X.** (co-first author), Han, B., Liu, K., Hu, D., & Liu, J. KV-Skill: Forging Expertise in the Model's Native Language. arXiv:2608.05475. Under review at ACL Rolling Review (Aug 2026 cycle). Converting a text skill lifts LiveMath (Qwen3.5-4B) 23.4→77.2; outperforms LoRA, prefix tuning, and soft prefixes in 7/8 matched-budget settings.
- Han, Z., **Zhang, X.** (co-first author), Guan, L., Hu, D., Liu, K., Chang, K., & Liu, J. Nothing Was Certified: Compositional Controllability for Fair Evaluation and Trust-Region Learning of Agent Systems. Under review at ICLR 2027.

OTHER PUBLICATIONS

- Liu, E., **Zhang, X.** (co-first author), Pan, P., Wang, Y., Liu, B., Zhang, L., & Mei, X. (2024, October). Agricultural technology gap affects climate change adaptation and mitigation based on patent data. *NPJ Climate and Atmospheric Science*, under review.
- Zhang, L., **Zhang, X.**, Gao, J., Lei, S., Yan, C., & Liu, E. (2024, September). An insight into the technological landscape and competitive situation of the biodegradable PBAT industry based on multidimensional patent analysis. *IEEE Transactions on Engineering Management*, under review.

SKILLS

- Distributed Training: **Deepspeed, DDP, FSDP**; RL infra: **TRL/veRL/OpenRLHF; vLLM; SGLang**
- Proficient in **Claude Code, Codex**, Python (PyTorch, Pandas, NumPy etc.), Java, R, SQL
- Skilled in using Linux servers with extensive experience, and proficient in shell scripting.
- GitHub: <https://github.com/shawnzhg>

INTERNSHIP

Baidu.com Times Technology (Beijing) Co., Ltd.

Beijing, China

Algorithm Intern

07/2024-10/2024

- Developed and optimized a speech recognition system using Transformer architecture to achieve high-precision audio-to-text transcription, particularly for dialect data
- Built Python-based web scraping tools to collect video data, extracted audio features, and used the features to improve speech-to-text accuracy through a customized encoder structure
- Applied dynamic time warping to align ASR and OCR outputs, improving text alignment accuracy for multi-language and dialect audio

HONORS AND AWARDS

- First-Class Merit-based Scholarship (Top 5%) 2023-2024
- First Prize, National College Student Market Research and Analysis Competition 2024
- Scientific Innovation Scholarship 2024
- Project Leader, Beijing University Student Innovation & Entrepreneurship Development Program 2024